Vietnam Portfolio Risk and Stress Report

FUESSV30 reference portfolio | VND100 billion shadow mandate | Information cutoff 30 September 2026

Decision

Retain the unhedged exposure under the current risk budget. Direct overlapping 10-day Expected Shortfall is below the 12% limit. Keep the beta 0.70 futures overlay as a contingency for a tighter absolute-risk mandate, subject to execution controls.

1-day 99% VaR	10-day 97.5% ES	Backtest	Beta to VN30
3.40%	**9.30%**	**2 / 250**	**0.99**

The one-day historical VaR limit passes with limited headroom. Direct overlapping 10-day ES is 9.30% and passes the 12% limit, while the square-root-of-time challenge is 12.30%. The direct method owns the limit because it preserves observed multi-day compounding and dependence.

Investment outcome

Mandate	Decision	Why
Current risk budget	Retain unhedged	Direct 10-day ES is 9.30% versus a 12% limit.
Tighter risk budget	Beta 0.70 contingency	The overlay reduces 99% historical VaR to 2.46% but increases tracking error.

This distinction is central. Hedging is not automatically superior; the correct action follows the mandate objective and approved risk budget.

Portfolio and data lineage

Portfolio construction

The model freezes the 30 September 2026 SSIAM VN30 ETF holdings and rescales them to a VND100 billion analytical mandate. Reported weights sum to 100.02% because of disclosure rounding; the analytical layer normalises them to 100.00% while retaining reported weights for audit.

Layer	Source	Use	Control
Holdings	SSIAM factsheet	Names, weights, sectors	Frozen cutoff and source page
Equities	VNDirect Finfo	Prices and traded values	Coverage and missing-quote audit
Benchmark	VNDirect Finfo	VN30 returns	Common cutoff and calendar
Futures	HNX	Multiplier and terms	Contract-selection gate
Validation	BIS MAR99	Traffic-light reference	Governance reference only

Short-history policy

TCX uses a Financials sector proxy excluding TCX in both windows, with a 1.33x peer-median volatility floor in the 500-day model. VPL uses an excluding-self Consumer Discretionary proxy in the 500-day model and its own history in the 250-day model. No substitution is silent.

Interpretation boundary

All historical results are current-weight replays. They answer how today's portfolio would have behaved under prior return environments. They are not reconstructed FUESSV30 fund returns and contain survivorship and composition bias.

Risk profile and model challenge

Measure	500-day	250-day	Interpretation
Annualised volatility	20.22%	20.63%	Stable long and recent estimate
EWMA volatility	14.23%	14.20%	Current regime is calmer
Historical VaR 99%	3.40%	3.25%	Empirical tail
Normal VaR 99%	2.87%	2.99%	Understates non-normal tail
Cornish-Fisher VaR 99%	4.17%	3.33%	Higher-moment challenge
Skewness	-0.80	-0.68	Negative left-tail asymmetry
Excess kurtosis	7.87	4.00	Normal distribution equals zero

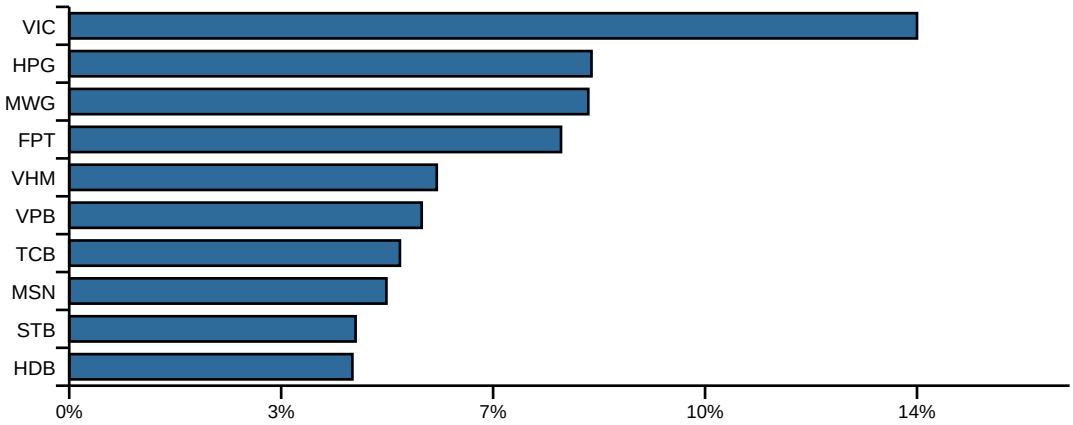
The distribution challenge changes the confidence placed in the result. A normal covariance model appears comfortably inside the 3.50% limit, historical VaR is near the boundary, and Cornish-Fisher VaR is higher. Historical ES and scenario loss therefore remain the main risk controls.

Regime and diversification

Average pairwise correlation is 0.37 over 500 days and 0.35 over the latest 60 days. The diversification ratio is 1.64. The current regime is calmer and is not exhibiting a broad correlation spike, but the historical left tail remains material.

Risk attribution and concentration

Top contributors to portfolio volatility



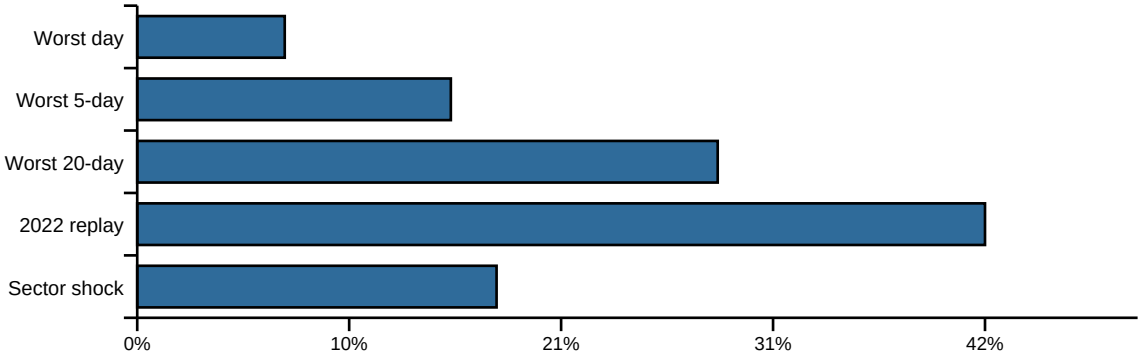
Name	Weight	Share of risk	Risk / weight
VIC	10.62%	13.54%	1.28x
HPG	8.72%	8.34%	0.96x
MWG	6.84%	8.29%	1.21x
FPT	8.62%	7.85%	0.91x
VHM	4.37%	5.87%	1.34x
VPB	4.72%	5.63%	1.19x
TCB	4.65%	5.28%	1.14x

VIC contributes 13.54% of portfolio volatility from a 10.62% weight. Real Estate contributes 21.85% of risk from a 16.88% weight, the clearest sector-level risk concentration.

Stress testing

Scenario	Loss	VND impact	Use
Worst historical day	-7.24%	VND -7.2bn	2020-03-09
Worst historical 5-day	-15.40%	VND -15.4bn	2025-04-02 to 2025-04-09
Worst historical 20-day	-28.50%	VND -28.5bn	2020-03-04 to 2020-03-31
2022 deleveraging replay	-41.62%	VND -41.6bn	Current-weight replay
Hypothetical sector shock	-17.64%	VND -17.6bn	Defined sector vector
Correlation-one 99% VaR	-4.86%	VND -4.9bn	Diversification failure

Scenario loss as a share of NAV



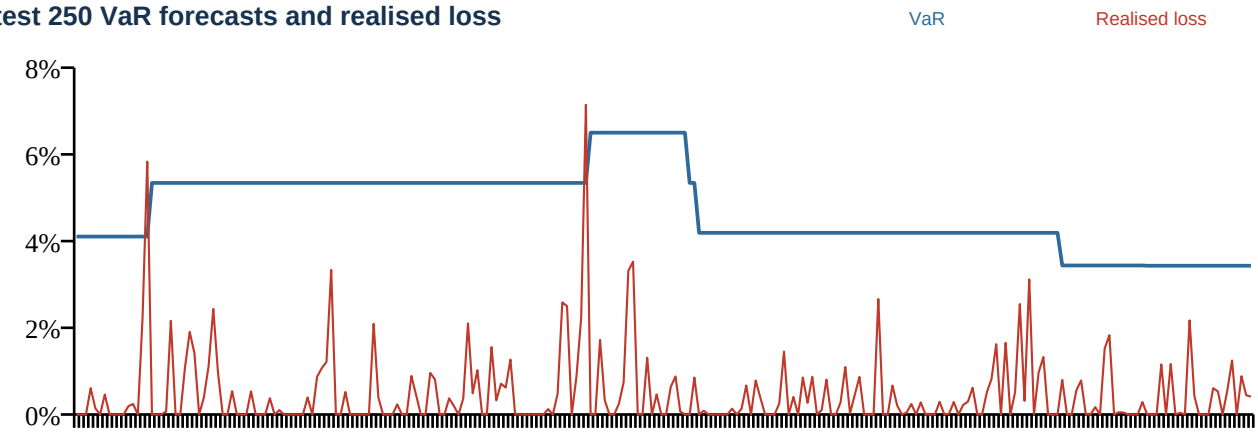
Reverse stress

A simplified beta-only reverse stress reaches a 15% NAV loss at an estimated VN30 shock of -15.15%. This is a threshold diagnostic, not a full nonlinear scenario.

Rolling diagnostic and validation

Control	Result	Interpretation
Forecast observations	250	Latest rolling forecasts
Exceptions	2	2 realised losses exceeded prior-day VaR
Traffic-light zone	GREEN	Green: 0 to 4 exceptions
Observed exception rate	0.80%	Expected rate is 1.00%
Kupiec POF p-value	0.742	Do not reject unconditional coverage at 5%
Christoffersen independence p-value	0.857	Do not reject independence at 5%
Conditional coverage p-value	0.932	Combined coverage diagnostic
Covariance PSD	PASS	Minimum eigenvalue 0.00e+00

Latest 250 VaR forecasts and realised loss



Validation conclusion

The rolling diagnostic passes the traffic-light, unconditional-coverage, independence and conditional-coverage tests. Each forecast uses only prior returns, but current 2026 holdings are replayed through history. This is not a production out-of-sample backtest of realised fund P&L.

Hedge analysis

Measure	Unhedged	Beta 0.70 hedge	Change
Annualised volatility	20.22%	14.40%	-5.82%
Historical VaR 99%	3.40%	2.46%	-0.94%
Historical ES 97.5%	4.16%	2.97%	-1.19%
Tracking error	2.35%	6.52%	4.17%
Beta	0.99	0.70	-0.29

The index-equivalent position is 152 short contracts at a VND 190.7 million notional per contract. Estimated VaR falls 27.55% and volatility falls 28.80%. Tracking error rises by 4.17%.

At the VSDC minimum initial margin input of 17.0%, the illustrated position requires about VND 4.9bn. A 1.5x liquidity buffer raises this to VND 7.4bn. Broker requirements may be higher; margin is not maximum economic loss.

Execution gates

Gate	Required before trade
Contract	Select maturity and use actual settlement price
Basis	Measure and limit cash-futures basis risk
Liquidity	Confirm broker margin above the VSDC minimum and fund the stress buffer
Roll	Define roll window, transaction cost and slippage
Governance	Approve tracking-error impact and rebalance tolerance

Governance, limitations and sources

Material limitations

- Frozen current weights introduce survivorship and composition bias in historical replays.

- Sector proxies cannot reproduce issuer-specific jumps; the peer-median volatility floor reduces but does not eliminate this limitation.

- Carry-forward creates zero returns on missing quote days and concentrates the move on the next quote.

- Direct 10-day ES uses overlapping observations, so sampling dependence reduces effective information.

- Square-root-of-time ES is retained only as a conservative challenge, not the limit-owning measure.

- Cornish-Fisher VaR can be unstable when sample higher moments are extreme.

- Liquidity applies a 50% volume haircut and 10% participation stress but still excludes price-limit queues and nonlinear market impact.

- The hedge includes a margin liquidity input but omits actual basis, maturity, variation-margin paths, roll, tax and execution costs.

Primary sources

SSIAM September 2026 ETF factsheet: https://ssiam.com.vn/upload/files/QuanLyQuy/ETF%20Monthly%20Report%20September%202026.pdf

SSIAM VN30 fund page: https://ssiam.com.vn/en/fund-information-VN30

HNX VN30 futures specification: https://www.vpdt.hnx.vn/vi-vn/huong-dan/chi-tiet-thu-tuc-36-65.html

VSDC initial margin notice effective 18 September 2026: https://vsdc.vn/vi/ad/200326

BIS Basel Framework MAR99: https://www.bis.org/committees/bcbs/basel-framework/standard/mar/99/inforce/2019-12-15/published/2019-12-15

VNDirect Finfo public API: https://api-finfo.vndirect.com.vn/v4

Reproducibility

The repository contains the frozen data snapshot, dependency-light Python risk engine, CSV outputs, a formula-linked decision workbook, model card and independent checks. Core risk statistics are generated by Python; decision inputs, hedge sizing and margin liquidity recalculate in Excel. Re-fetching data creates a new evidence set and should be versioned as a new model run.